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Testing the Conditions for Acquiring Intuitive Expertise in Judgment: Evidence from a Study of NCAA Basketball Tournament Predictions

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Evidence from psychological studies of both novice and expert forecasters often show that people are overconfident in their ability to make accurate judgmental predictions about future events. In contrast, evidence from much cognitive engineering research appears to show that experienced performers are reasonably well adapted to their task environments and do not display many of the cognitive biases, such as overconfidence, evident in the psychological literature. We present the results of a study providing evidence for both views, and more importantly, for the conditions under which experienced forecasters will exhibit high levels of calibration (little overconfidence) versus poorer calibration and thus overconfidence. In short, good calibration resulted when forecasters were required to draw solely upon high validity base-rate information gained through many years of repetitive exposures to a prediction task, yet displayed overconfidence when they were also provided case-specific information with high salience and interest but little or no predictive validity.

INTRODUCTION

Evidence from psychological studies of both novice and expert forecasters often show that people are overconfident in their ability to make accurate predictions about future events (Tetlock, 2006). Yet, another body of research performed by cognitive engineering researchers often finds that experienced performers display many fewer of the cognitive biases than basic psychological research would lead us to expect (Klein, 1999; Mosier & Fischer, 2010). Cognitive engineering research and applications focusing on improving the accuracy of predictions and forecasts in a wide variety of domains requires knowledge of the conditions under which people can be expected to become well adapted and thus truly expert, versus the conditions under which their performance will suffer due to biases and cognitive limitations, so that interventions such as technology and training can be targeted to the latter.

In forecasting tasks, uncertainty places a ceiling on predictive accuracy, so perfectly accurate performance is impossible when uncertainty is present. In these situations, the best a forecaster can aim for is to provide forecasts that are well calibrated, in the sense that if a forecaster states that event X is expected to occur with probability P, that, over time and under similar conditions, the relative frequency with which X occurs is approximately equal to P. Forecasted P values that are consistently higher than observed relative frequencies are taken to indicate an overconfidence bias (Brenner, Koehler, Liberman, & Tversky, 1996).

Recently, and in an important paper, Kahneman and Klein (2009) proposed a resolution to the body of conflicting findings on the prevalence of overconfidence by stating that people making repeated intuitive judgmental predictions in environments of high validity (i.e., predictability) and rapid, unambiguous feedback are likely to become truly expert in the sense of being well calibrated. In contrast, these authors stated that overconfidence is likely to result when predictive success is due more to chance than skill: lucky forecasts will contribute to an illusion of skill and thereby to overconfidence (Arkes, 2001).

We seized upon the 2012 NCAA Mens Basketball tournament to provide an experimental test of Kahneman and Klein's proposal, due in large part to having available a large number of highly experienced participants (viewers of many past tournament games) in a university student population.

NCAA TOURNAMENT

The National Collegiate Athletic Association (NCAA) is the governing body over the athletic programs of most American universities. Every March they organize the NCAA Men's Division I Basketball Championship Tournament, a 68-team single elimination playoff. Teams are invited to the tournament either through a selection committee or because of automatic bids for winning their own conference tournament. To balance the tournament, each team is given a seed corresponding to their ranking between 1 and 16. This ranking is based on each team's regular season performance.

A team with a seed of 1 represents a team that the committee perceives to be stronger than a team with a seed of 16. An early round of the NCAA Tournament has 32 games between 64 teams, with each seed being represented by four different teams. In this round, the strongest team, a team seeded first, plays the weakest team, a team seeded sixteenth, the second strongest team, a team seeded second, plays the second weakest team, a team seeded fifteenth, and so on. For our study we focus on this round of the NCAA Tournament. The history of the winning percentages of the lower seeded team is shown in Table 1. As the difference in seed numbers decreases, as does the relative frequency of the more highly seeded team winning.

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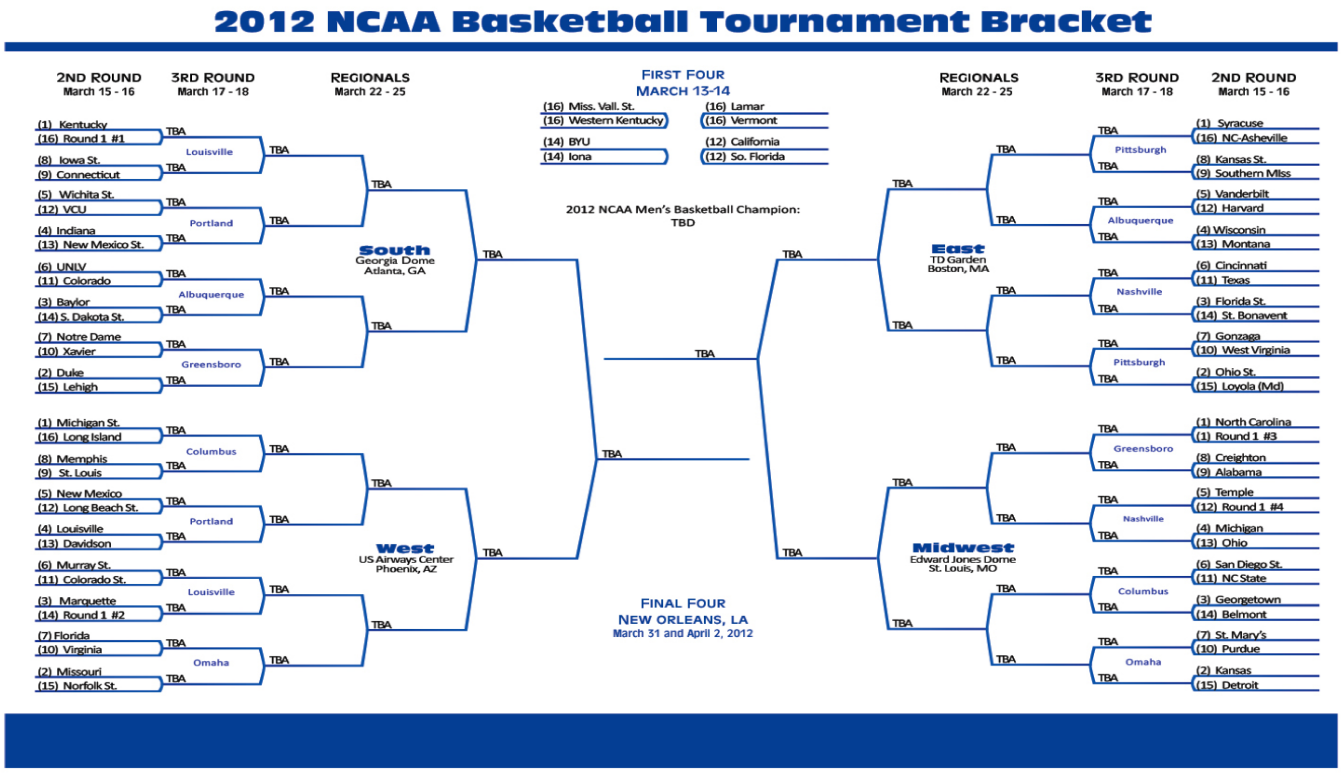


Figure 1: The tournament bracket subjects were presented with in the conditions where team and seed match-up information were provided.

Match-up	Historical Percentages
8 vs 9	50.8% (N = 132)
7 vs 10	59.8% (N = 132)
6 vs 11	67.2% (N = 128)
5 vs 12	69.5% (N = 128)
4 vs 13	78.7% (N = 108)
3 vs 14	85.2% (N = 108)
2 vs 15	96.3% (N = 108)
1 vs 16	100.0% (N = 108)

Table 1: The percent of games the lower seeded team has won in previous NCAA Tournaments.

METHOD

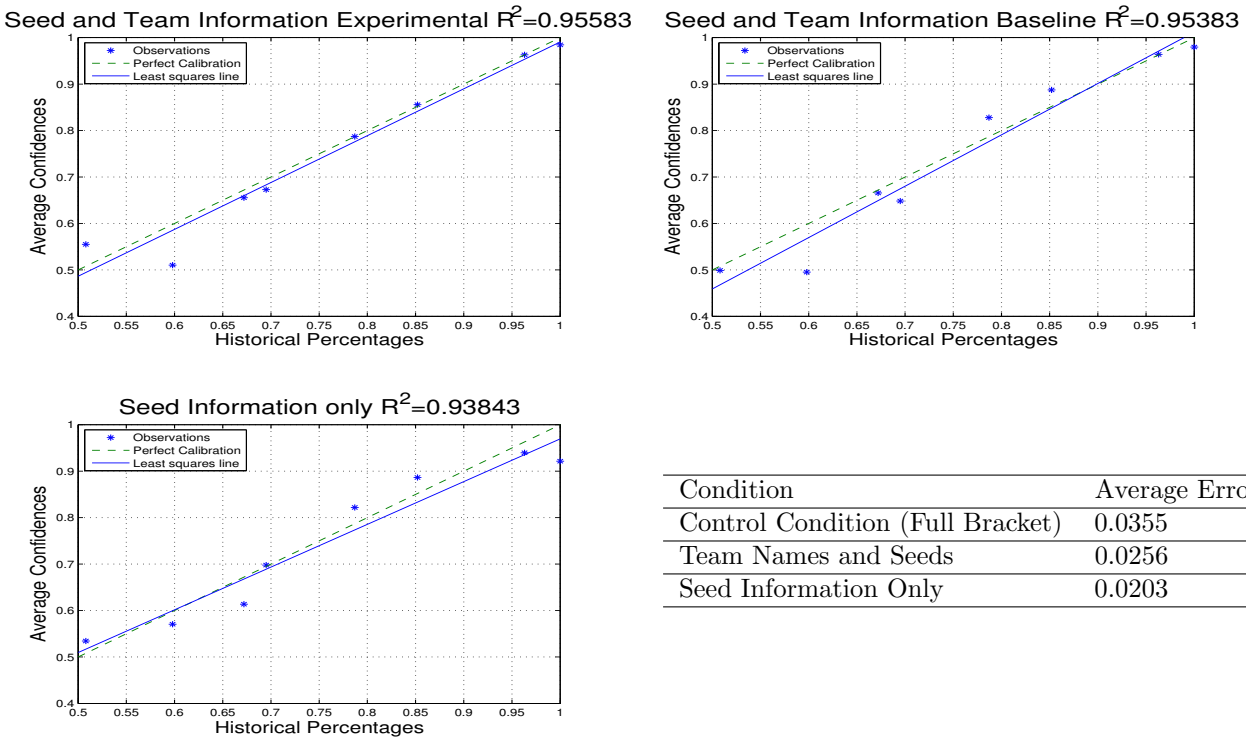
We chose this domain because the tournament is watched by millions of people each year and we believed our subject pool would be eager to work on a project in this domain. We recruited forty-six (N=46) total subjects for this experiment, paying each $25 for their participation. To entice more subjects, the two subjects with the best performance in each experimental group were awarded an additional $100. Groups were comprised primarily of college undergraduate students. Data collection was conducted over a five-day period, with the first experimental group's data being collected after all season games were completed and 2 days prior to the NCAA announcing the team match-ups. A pre-test was given before each subject began the experiment. The test consisted of oral questions that demonstrated that they had basic knowledge of both current and prior years' NCAA Tournaments.

A paper-based trial was then given to provide familiarity with the style of question to be used in the experiment. The trial questions involved a domain other than the NCAA Tournament, but followed the format of the questions to be answered in the experiment and allowed the subjects to ask the experimenters questions.

We then gave the subjects the experimental questions. In the first condition (N=22), subjects were asked to make predictions based on solely the seeding match-up information as in Figure 3. Since this data was collected before the seeds for specific teams were released, subjects had no knowledge of which teams corresponded to which seeds. In addition, since each seed represents four teams (i.e. North Carolina, Michigan State, Syracuse, and Kentucky could all have seeds of 1), then no team is uniquely identified by a seed, thus subjects could not guess as to which teams would correspond to the match-ups in question. The experimental questions included the following:

1. Will the 1-seeded team defeat the 16-seeded team? Yes/No

2. What is your confidence in your prediction for question 1 on a scale of 50% to 100% with 50% indicating



Condition	Average Error
Control Condition (Full Bracket)	0.0355
Team Names and Seeds	0.0256
Seed Information Only	0.0203

Figure 2: The dotted line represents perfect calibration. (Top Left) The average predictions made by subjects that did condition 1 (no team information) and then condition 2 (team and seed information). (Top Right) The average predictions made by subjects that did condition 2 only. (Bottom Left) The average predictions made by subjects in condition 1. (Bottom Right) The average error, judged against historical data, found in each condition per question.

you are as confident as when flipping a fair coin and 100% indicating that you are certain.

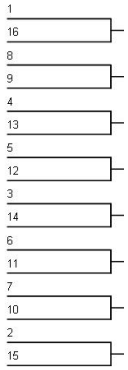


Figure 3: The bracket used in condition 1 showing only the seeds.

After the seeding was announced, we began the second condition. The original subjects (N=22), who had previously made predictions solely on seed number information, and a group of 24 subjects new to the study were now asked to answer similar questions as before but now, after the NCAA had announced seeding information, in the form "Will 1-seeded Kentucky defeat 16-seeded Western Kentucky? Yes/No," with the names and seeds of the teams changing according to the released NCAA seedings and teams. This allowed for the collection of both an experimental condition that contained subjects that provided predictions when team names were unknown and a control condition that simply completed the bracket in the normal fashion. We chose to use the team seeds as our base rate information because data for it is easily attainable and scenarios have been repeated a minimum of 108 times, as in the one versus sixteen seed matchup.

RESULTS

We will first present the results for the three conditions and calibration curves against historical data. On our first pass of the data, we saw that subjects were well calibrated with the historical percentages shown by Figure 2 and the average error, or deviation for each matchup was fairly small. In all three conditions, some fraction of subjects predicted an "upset" (team with the higher seed defeating a team with a lower seed). This occurred in 20% of the match-ups predicted in the control condition, 16% of the match-ups predicted with team and seed information, and 12% of the match-ups predicted with just seed information. To create calibration curves against histori-

cal data, we recoded subject confidence values from their provided P to (1-P). So, if a subject indicated that she was 70% confident that a 12 seeded team would defeat a 5 seeded team, this value was recoded as a 30% confidence that the "favored" team (team with the lower seed) would win. Once completed, these predictions were averaged with the subjects that predicted favorites.

By doing this recoding and aggregation of confidence values, we made it possible for the data to show good calibration to historical base-rates in one of two different ways. The first possibility was subjects simply had a high level of calibration resulting from predictions aligned with the historical base-rates given in Table 1. The second possibility was an apparent high level of calibration resulting from the type of overconfidence often observed in forecasting studies combined with a irrational and empirically unjustifiable number of upset predictions, which when averaged with any overconfidence when predicting favorites, was actually two types of biases canceling out. These two biases were overconfidence in the prediction of favorites and a representativeness bias (Kahneman & Tversky, 1972) in the selection of upsets. Subjects knew that some fraction of the games would result in an upset, so they strove to make their overall predictions representative by including both upset and favorite predictions. Though upsets are likely to occur, it is not rationally defensible for any particular individual game to result in an upset, as no algorithm exists that exceeds the prediction ability of predicting the team with the lower NCAA seed for every game. Our further analysis will show that when subjects predicted upsets they did not significantly improve their predictive accuracy.

This consideration prompted our interest in creating calibration curves that excluded upset predictions (Figure 4). In the control condition, we find that subjects are significantly overconfident in the favored team winning with $p < 0.001$ for all match-ups where the favored team wins less than 95% of the time based on historical data (Figure 4 top right). In the experimental condition, when subjects are provided only with the seed information, they are much more calibrated to historical percentages with $p < .05$ only in the "8 vs 9" seed match-up and "1 vs 16," (Figure 4 bottom left). When the experimental condition is provided with the team information as well, they too exhibit significant overconfidence with $p < 0.01$ for all match-ups where the favored team wins less than 85% of the time based on historical data (Figure 4 top left).

To investigate if subjects performed significantly better by attempting to predict upsets, and thus knew information not represented in the model based on historical percentages, we used a Brier score. Brier score is one method used to assess the accuracy of predictions (Glenn & Caskey, 1950). Brier score is a strictly proper scoring rule that measures the amount of error a judge or forecaster has in his or her prediction. To understand the way a strictly proper scoring rule behaves, consider assessing the probability distribution for a set of mutually exclusive exhaustive events, so

$$E = \{e_1, e_2, ..., e_n\}$$

$$\sum_{i=1}^{n} p(e_i) = 1,$$

where e_i is event i and $p(e_i)$ is the probability of event i occurring. Now, an assessor, Y, has their private beliefs about the probabilities for the distribution for event E. We call this vector of their private beliefs P, where each element ranges from [0, 1] and the sum of the vector is 1. Now when asked to publicly assess their beliefs for the distribution of event E, the forecaster provides a set of responses R, ranging for [0, 1] and summing to 1. These responses are their public responses. A strictly proper scoring rule is a rule that maximizes the forecaster's score when R = P (Bickel, 2007). The Brier scoring rule is a transformation from the quadratic scoring rule and is represented as

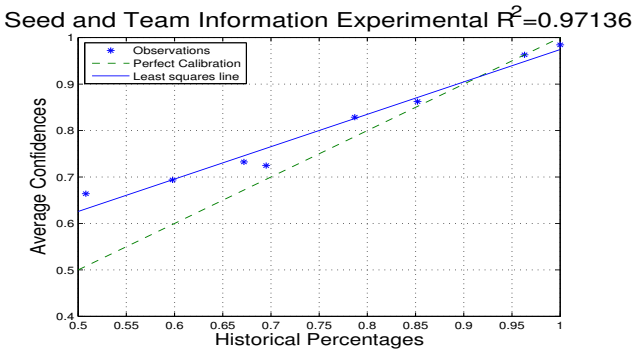
$$BS = \frac{1}{N}\sum_{t=1}^{N}(f_t - o_t)^2.$$

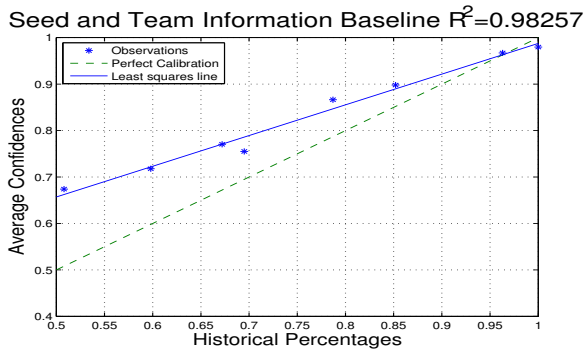
N represents the number of forecasting problems, f_t represents the probability that the judge forecasts, and o_t is the actual outcome of the event, 1 if the event does happen and a 0 if it does not. We found that the Brier scores for the control condition decreased from 0.2349 to 0.2149 when accounting for upsets and decreased from 0.2247 to 0.2090 for the experimental condition when accounting for upsets (Figure 4). Neither of those decreases were significant.

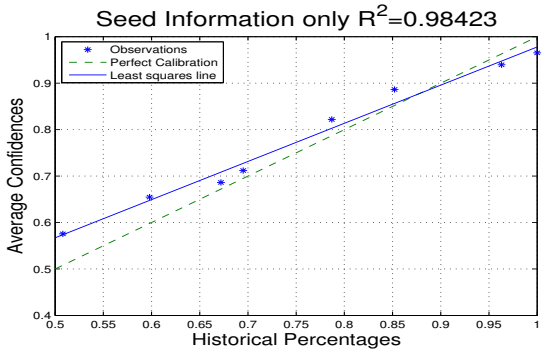
DISCUSSION

We have shown evidence that supports the views of both psychological and cognitive engineering studies on how experienced forecasters make calibrated predictions about future events. When subjects were provided with both the team name and seed match-up information, they demonstrated the traditional overconfidence exhibited by subjects on forecasting problems, as in previous psychological studies. We had to adjust for matches subjects predicted as upsets to view this overconfidence, but the lack of significant Brier score decreasing shows that these predictions of upsets did not significantly improve their accuracy. When subjects only had seed match-up information to make their judgment, we showed that they can be well calibrated. Subjects had experience with the NCAA Tournament and the unambiguous feedback they received over the years allowed them to use this expertise to behave in a well calibrated fashion. This supports the resolution of Kahneman and Klein (2009) that in a domain with high validity and rapid feedback, subjects are likely to become well calibrated. Though the added information of team names was salient, it had little or no predictive validity, and instead made them overconfident.

The goal of this research was to investigate how the effects of case-specific information influenced the confidence of subject forecasts. The results indicate that withholding







Condition	Brier Score w/ Upsets
Control (Full Bracket)	0.2349
Team Names and Seeds	0.2247

Condition	Brier Score w/o Upsets
Control (Full Bracket)	0.2149
Team Names and Seeds	0.2090

Figure 4: Graphs show the calibration without upset predictions. The dotted line represents perfect calibration. (Top Left) The average predictions made by subjects that did condition 1 (no team information) and then condition 2 (team and seed information). (Top Right) The average predictions made by subjects that did condition 2 only. (Bottom Left) The average predictions made by subjects in condition 1. (Bottom Right) The mean Brier scores, judged against actual outcome, for conditions with team names with and without upset predictions.

case-specific information can eliminate much of the overconfidence bias commonly displayed by subjects performing a forecasting task, in cases where base-rate information is available and diagnostic in predicting outcomes, and case-specific information has little predictive value. Though our experiment is limited to the NCAA Tournament, the method used of hiding case-specific information could be used in forecasting political and regional events by intelligence analyst when they have been exposed to base rate data from experience.

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